

Zhengni Hu

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PERSONAL INFORMATION

- Date of Birth Oct. 01, 1996.
- Place of Birth Changsha, Hunan, China.
- Citizenship China.

RESEARCH INTERESTS

Nonlinear Partial Differential Equations and Geometric Analysis

- Nonlinear partial differential equations arising in geometry and mathematical physics, including mean field equations, Liouville-type systems, super-Liouville equations, and related models
- Qualitative and geometric analysis of solutions to nonlinear PDEs, with a focus on singular phenomena such as blow-up and concentration
- Variational methods, finite-dimensional reduction and “critical point at infinity” techniques for nonlinear PDEs

WORK EXPERIENCE

Wen-Tsun Wu Assistant Professorship (Postdoctoral Fellow)

Shanghai Jiao Tong University

Dec. 2024 – present

(Anticipated end date: Dec. 2026 or Mar. 2027, flexible)

Advisors: Prof. Miaomiao Zhu & Prof. Tongrui Wang

EDUCATION

- **2021–2024** Ph.D. in Mathematics, Justus Liebig University Giessen, Germany.
 - Dissertation: *Blow-up Solutions for Mean Field Equations and Toda Systems on Riemann Surfaces*
 - Advisors: Prof. Thomas Bartsch & Prof. Mohameden Ahmedou
- **2018–2021** M.Sc. in Mathematics, Peking University, Beijing, China.
 - Thesis: *On Elliptic Solutions of a Generalized Monge–Ampère Equation*
 - Supervisor: Prof. Zhifei Zhang
- **2014–2018** B.Sc. in Mathematics and Applied Mathematics, Beijing Normal University, Beijing, China.
 - Thesis: *Compact Embedding Between Besov Spaces and Triebel–Lizorkin Spaces on Metric Spaces* (evaluated as an excellent bachelor thesis)
 - Supervisor: Prof. Wen Yuan

PUBLICATIONS

1. Blow-up solutions for the steady state of the Keller–Segel system on Riemann surfaces. M. Ahmedou, T. Bartsch, and Z. Hu, *Calc. Var. Partial Differential Equations*, 2026. DOI: 10.1007/s00526-026-03313-5
2. A degree-counting formula for a Keller–Segel equation on a surface with boundary. M. Ahmedou, Z. Hu and H. Wang. *Commun. Contemp. Math.*, 2026. DOI:10.1142/S0219199726500392

3. Blow-up solutions for general Toda systems on Riemann surfaces.
Z. Hu and M. Zhu, *J. Differential Equations*, 2026. DOI: 10.1016/j.jde.2026.114145
4. The Morse property of limit functions appearing in mean field equations on surfaces with boundary.
Z. Hu and T. Bartsch, *J. Geom. Anal.*, 2024. DOI: 10.1007/s12220-024-01664-z

PREPRINTS

1. Blow-up solutions for mean field equations with Neumann boundary conditions on Riemann surfaces. with T. Bartsch and M. Ahmedou. arXiv:2408.16917 v2 .
2. Partial blow-up phenomena in the $SU(3)$ Toda system on Riemann surfaces. with T. Bartsch and M. Ahmedou. arXiv:2408.17372.
3. Blow-up solutions for mean field equations with non-quantized singularities on Riemann surfaces with boundary. with M. Ahmedou and M. Zhu. arXiv:2602.04790.

TEACHING

Mathematical Methods in Physics
Lecturer, Shanghai Jiao Tong University

Fall 2025 – Spring 2026

TALKS

- Seminar on PDEs, Chongqing Jiao Tong University, China
May 2026
- Seminar on Analysis and Geometry, Justus Liebig University Giessen, Germany
Feb. 2026
- Seminar on Analysis and Geometry, Shanghai Jiao Tong University, China
Mar. 2025
- The 1st International Young Scholars Forum, Yunnan Normal University, China
Jan. 2025
- Seminar on Analysis and Geometry, Justus Liebig University Giessen, Germany
Jul. 2024
- Seminar on Analysis and Geometry, Justus Liebig University Giessen, Germany
Jun. 2024
- VIII Symposium on Nonlinear Analysis, Nicolaus Copernicus University, Poland
Jun. 2024
- Summer School on Calculus of Variations: Variational and Geometrical Methods, University of Warsaw, Poland
Jun. 2024
- Atlantic Conference on Nonlinear Partial Differential Equations, Instituto Superior Técnico, Portugal (Poster Presentation)
Nov. 2023

GRANTS AND FELLOWSHIPS

- *China Scholarship Council (CSC) Fellowship* 2021–2024
No. 202106010046.

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